

Socho tumhare paas ek dabba hai jiska naam hai **chocolates**. Us dabbe ke andar tumne 5 chocolates rakhi:

```
var chocolates = 5;
```

Ab dabbe me 5 chocolates hain 

Phir kisi ne tumhe aur 5 chocolates de di, aur tumne wo bhi same dabbe me daal di:

```
chocolates = chocolates + 5;
```

Yaha kya hua?

Right side pe JavaScript ne bola: “dabbe me pehle se 5 hain, usme 5 aur add karo → 10”

Phir wo 10 wapas dabbe me store ho gaya.

Ab dabbe me 10 chocolates ho gayi 

Agar tum aur chocolates add karte ho:

```
chocolates = chocolates + 2;
```

To ab total ho gaya 12.

Important baat ye hai ki:

```
chocolates = chocolates + 5
```

Ye maths ka equation nahi hai, balki ek **instruction** hai —

“pehle dabbe ki value uthao, usme 5 add karo, aur result wapas same dabbe me daal do.”

A **variable** in JavaScript is simply a name that stores a value, so you don't have to write the value again and again. For example:

```
var x = 100;  
ellipse(x + 50, 200, 50, 50);
```

Here, `x` stores the value **100**, and when we write `x + 50`, JavaScript uses the value inside `x` and adds 50, so the ellipse is drawn at position **150**. The value of `x` itself does not change — we are just using it in a calculation.

The **need of variables** is to make programs easier, flexible, and reusable. Instead of writing fixed numbers everywhere, we store them in variables and use them wherever needed. If we want to change something, we only change the variable once instead of changing it in many places. For example, if `x` controls position, changing `x = 200` will automatically move the shape without modifying every line. Variables also help in calculations, animations, and dynamic behavior, which is very important in graphics like Processing.js.

With Colouring Functions...

```
var r = 100;
```

```
var g = 50;
```

```
var b = 150;
```

```
fill(r + 50, g + 20, b - 50);
```

```
ellipse(200, 200, 100, 100);
```

👉 Samjho:

- $r + 50 \rightarrow 100 + 50 = 150$
- $g + 20 \rightarrow 50 + 20 = 70$
- $b - 50 \rightarrow 150 - 50 = 100$

👉 Final color:

```
fill(150, 70, 100);
```

✓ Variable same raha

✓ Use karte waqt calculation hui

Functions

Socho tumhare paas chocolates ka ek dabba hai. Kabhi tum chocolates add karte ho, kabhi double karte ho, kabhi friends me distribute karte ho. Agar har baar tum ye same kaam manually karoge, to baar-baar same steps repeat karne padenge — thoda boring aur confusing ho jayega.

Is problem ka solution hai **function**.

👉 Function ko simple words me samjho:

function ek fixed kaam (task) hota hai jo baar-baar use kiya ja sakta hai.

Socho ek **machine** hai:

- tum usme chocolates daalte ho
- machine ek specific kaam karti hai (jaise +5 add karna)
- aur result wapas deti hai

Ab jab bhi tumhe chocolates me +5 add karna ho, tumhe manually nahi karna padega — bas machine use karo.

👉 Iska fayda:

- baar-baar same kaam repeat nahi karna
- code clean aur simple hota hai
- easily reuse kar sakte ho
- mistake kam hoti hai

A **function in JavaScript** is a reusable block of code designed to perform a specific task. It can take inputs (called parameters), process them, and return an output. Functions help in organizing code, improving reusability, and reducing repetition.

Ways to declare a function...

1. Function Declaration (Named Function)

This is the most common way.

```
function add(a, b) {  
    return a + b;  
}
```

2. Function Expression (Using **var**)

```
var add = function(a, b) {  
    return a + b;  
};
```

3. Arrow Function (Modern JavaScript)

```
var add = (a, b) => {  
    return a + b;  
};
```